

Multiple Choice (10 marks)

1. If a random variable X is normally distributed with a mean of 118 and a standard deviation of 11, what Z-scores correspond to raw scores of 115, 134, and 99?
 - (a) -.27, 1.45, 1.73
 - (b) -.27, -1.45, -1.73
 - (c) -2.18, 0.15, -1.82
 - (d) 0, 1.5, -2
 - (e) .27, 1.45, 1.73
2. This biome contains plants whose roots cannot go deep due to the presence of a permafrost.
 - (a) Savanna
 - (b) Taiga
 - (c) Tundra
 - (d) Chaparral
 - (e) Alpine
3. What are the major forces of evolution?
 - (a) reproduction, migration, mutation, natural selection
 - (b) mutation, genetic drift, migration, genetic hitchhiking
 - (c) mutation, genetic drift, migration, random mating
 - (d) mutation, gene flow, migration, natural selection
 - (e) adaptation, genetic drift, migration, natural selection
4. Which group is composed entirely of individuals who maintained that species are fixed (i.e., unchanging)?
 - (a) Aristotle, Darwin, and Lamarck
 - (b) Aristotle, Lyell, and Darwin
 - (c) Aristotle, Linnaeus, and Cuvier
 - (d) Aristotle, Cuvier, and Lamarck
 - (e) Linnaeus, Darwin, and Lamarck
5. Fungi participate in each of the following EXCEPT
 - (a) photosynthesis to produce glucose

- (b) fermentation to produce alcohol
- (c) association with humans to produce ringworm
- (d) association with the roots of plants to form mycorrhizae

True / False (10 marks)

Answer True or False

- 6. True or False: Gene flow between populations increases the likelihood of speciation in the metapopulation.
- 7. True or False: Maintaining the mosquito population at the highest point of its logistic curve will ensure long-term population stability.
- 8. True or False: The extra Y chromosome can decrease a male's aggression.
- 9. True or False: Commensalism is not considered an interspecies interaction because it does not involve any relationship between different species.
- 10. True or False: The conversion of pyruvic acid to acetyl CoA is not an example of hydrolysis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the methods used to estimate the age of rock formations, focusing on the role of radioactive elements and fossil content in this process.
- 12. Discuss the concepts of incomplete dominance and epistasis in genetics, highlighting their differences and similarities in terms of allele interactions and gene relationships.

End of Paper